

Problem 18

Two chess players play one game. If the first player wins, this is event A ; if the second player wins, this is event B . What event should be added to this set of events to obtain a complete system of events?

Solution

A complete system of events (partition) must satisfy:

1. All events are mutually exclusive (disjoint)
2. The union of all events equals the sample space S

Step 1: Analyze the Given Events

Event A : First player wins

Event B : Second player wins

Step 2: Check if A and B are Mutually Exclusive

In a single chess game, only one player can win (assuming no draws initially), so:

$$A \cap B = \emptyset$$

The events A and B are mutually exclusive.

Step 3: Check if A and B are Exhaustive

For A and B to form a complete system alone, we would need:

$$A + B = S$$

However, in chess, there is a third possible outcome: the game can end in a **draw** (remí).
Therefore:

$$A + B \neq S$$

The events A and B do NOT cover all possible outcomes.

Step 4: Identify the Missing Event

The missing event is:

Event C : The game ends in a draw (remí)

This can be expressed as:

$$C = \overline{A + B} = \bar{A}\bar{B}$$

This represents the event where neither player wins.

Step 5: Verify the Complete System

Now we have three events: A , B , and C .

Mutual Exclusivity

Check $A \cap B$:

$$A \cap B = \emptyset \quad (\text{only one player can win})$$

Check $A \cap C$:

$$A \cap C = A \cap \bar{A}\bar{B} = A\bar{A}\bar{B} = \emptyset$$

Check $B \cap C$:

$$B \cap C = B \cap \bar{A}\bar{B} = \bar{A}B\bar{B} = \emptyset$$

All three events are pairwise disjoint.

Exhaustivity

$$A + B + C = A + B + \overline{A + B}$$

Using the complement law:

$$= (A + B) + \overline{(A + B)} = S$$

The union equals the sample space.

Step 6: Interpretation

The complete system of events describes all possible outcomes of the chess game:

- **Event A:** First player wins (1-0)
- **Event B:** Second player wins (0-1)
- **Event C:** Draw ($\frac{1}{2}$ - $\frac{1}{2}$)

Every chess game must result in exactly one of these three outcomes.

Alternative Notation

The missing event can also be denoted as:

$$C = \bar{A}B + A\bar{B} + \bar{A}\bar{B}$$

Wait, that's not correct. Let me reconsider.

Actually, the events should be defined more carefully: - A : First player wins - B : Second player wins - C : Draw

These are naturally disjoint outcomes, and C is simply the complement of $A \cup B$:

$$C = \overline{A + B}$$

Probabilistic Interpretation

If we denote: - $P(A) = p_1$ (probability first player wins) - $P(B) = p_2$ (probability second player wins) - $P(C) = p_3$ (probability of draw)

Then for a complete system:

$$p_1 + p_2 + p_3 = 1$$

$$P(A) + P(B) + P(C) = 1$$

Example with Specific Probabilities

Suppose: - $P(A) = 0.4$ (first player wins 40- $P(B) = 0.3$ (second player wins 30- $P(C) = 0.3$ (draw occurs 30

Then $0.4 + 0.3 + 0.3 = 1.0$

Final Answer

$$C = \overline{A + B} = \text{Draw (remí)}$$

The event that should be added is: ****The game ends in a draw****.

Complete system of events:

- A - First player wins
- B - Second player wins
- $C = \overline{A + B}$ - Draw

Verification:

$$\text{Mutual exclusivity: } A \cap B = A \cap C = B \cap C = \emptyset$$

$$\text{Exhaustivity: } A + B + C = S$$

Note: In chess terminology, the three possible results are:

- 1-0 (white wins) $\rightarrow$ Event A (if first player is white)
- 0-1 (black wins) $\rightarrow$ Event B (if second player is black)
- $\frac{1}{2}-\frac{1}{2}$ (draw) $\rightarrow$ Event C

These three outcomes partition the sample space of all possible chess game results.